



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

PAGE WAVE

The domain of the Project:
G20 JAVA APPILICATIONS

Team Mentors (and their designation):
CHANDHANA TAGADI
PRUTHVI RAJ

Team Members:
Mr. G.S.V.S HITISH KUMAR

Period of the project

December 2025 to February 2026



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Declaration

The project titled “PAGE WAVE” has been mentored by T. CHANDHANA & PRUTHVI RAJ, organised by SURE Trust, from December 2025 to February 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. G.S.V.S Hitish kumar

Mentor's Name

T. Chandhana—Company Name

Mentor's Name

Pruthviraj —Company Name

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

Project Title: PAGE WAVE – Document Printing Service

Domain: G20 Java Applications

Duration: December 2025 – February 2026

Organization: SURE Trust

Team Members: Mr. G.S.V.S HITISH KUMAR

Mentors: T. CHANDHANA & PRUTHVI RAJ.

PAGE WAVE is a web-based Document Printing Service developed using Java and Spring Boot. The project was undertaken as part of the SURE Trust skill development program, aimed at empowering educated unemployed rural youth with industry-relevant practical experience.

The application addresses a persistent problem faced in educational institutions and small offices—manual, unorganized printing systems that rely on WhatsApp, email, or USB drives. These methods lack transparency, create long queues, and result in paper wastage and mismanagement.

PAGE WAVE digitizes the entire printing workflow. Students can upload documents, select print preferences (color, copies, duplex, page range), and track their job status in real time. Administrators gain access to a centralized dashboard to monitor all print jobs, filter requests, and update statuses. The system assigns a unique Job ID to each request, ensuring accountability and traceability.

Objectives

The primary objectives of this project were:

1. To design and develop a secure, role-based web application for print job management.
2. To implement user authentication and authorization using Spring Security.
3. To enable file upload functionality with format and size validation.
4. To provide customizable print options and real-time job tracking.
5. To build an admin interface for centralized job monitoring and control.
6. To estimate printing costs based on page count and color selection.



Introduction

The PAGE WAVE project was developed under the SURE Trust program to address the disorganized printing systems prevalent in educational institutions. Currently, students rely on WhatsApp, email, or USB drives to submit print jobs, with no tracking or queue management. This leads to long wait times, lost files, duplicate prints, and unequal access.

This project aims to build a web-based document printing service with role-based authentication, file upload, print job tracking, and an admin dashboard. The scope includes user management, job submission, status updates, and cost estimation. Limitations include no real printer integration, no payment gateway, basic UI, and local deployment only.

The innovation lies in providing a fair, first-come-first-served digital queue using zero-cost open-source technologies, specifically designed for rural and semi-urban institutions with limited infrastructure.

Key Points:

- Problem: Manual printing system causes inefficiency and unfair access
- Goal: Digitize print job submission and tracking
- Scope: Authentication, file upload, job status, admin panel, cost estimation
- Limitations: No hardware integration, no payments, basic UI
- Innovation: Free, fair digital queue for resource-constrained settings



Project Objectives

Primary Objectives

1. **To develop a role-based authentication system** – Implement secure login and registration for Students and Administrators using Spring Security with BCrypt password encryption.
2. **To enable document upload functionality** – Allow users to upload PDF, DOCX, and JPG files with validation for file type (max 10MB) and store them securely on the server.
3. **To provide customizable print settings** – Let users select color mode (B&W/Color), number of copies (1-10), page range (All/Custom/Single), and layout (Single-sided/Double-sided).
4. **To implement print job tracking** – Generate a unique Job ID for each request and maintain real-time status updates (Pending → Completed/Cancelled).
5. **To build an admin dashboard** – Provide administrators with a centralized view to monitor all print jobs, filter by user/date/status, and update job status.
6. **To calculate estimated printing cost** – Develop a pricing formula based on page count and color selection (₹2/page B&W, ₹5/page Color).

Secondary Objectives

7. **To ensure proper validation and error handling** – Prevent duplicate submissions, validate user inputs, and display meaningful error messages.
8. **To maintain session management** – Implement login session timeout and secure logout functionality.
9. **To write clean, documented code** – Maintain readable code with comments for future reference and collaboration.
10. **To use version control** – Manage project code on GitHub with regular commits.

Expected Outcomes

S.No	Deliverable	Description
1.	Functional Web Application	A working document printing service accessible via web browser.
2.	User Authentication Module	Login/Register pages with role-based redirection
3.	File Upload System	Document upload with validation and local storage



Methodology and Results

Technology Stack

Backend: Java 17, Spring Boot, Spring Security, Spring Data JPA

Frontend: HTML, CSS, JavaScript, Thymeleaf

Database: MySQL

Tools: IntelliJ IDEA, Postman, Git, MySQL Workbench

Build Tool: Maven

Project Architecture

Pattern: Model-View-Controller (MVC)

Browser → Controller → Service → Repository → Database

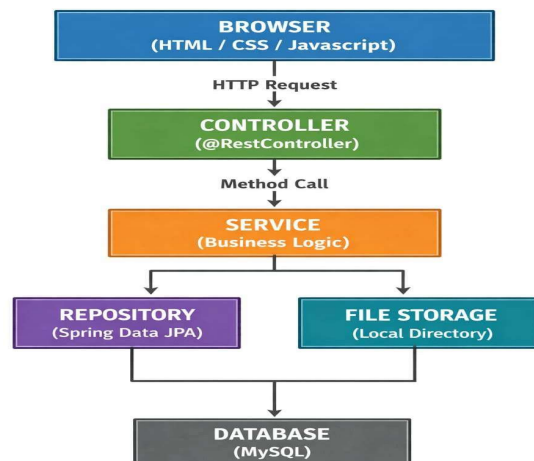


File Storage (Local)

Flow:

- Controller handles requests
- Service contains business logic
- Repository interacts with MySQL using JPA
- Files saved locally, paths stored in DB
- Thymeleaf renders HTML views

Packages: controller, service, repository, entity, config





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Screenshots (Summary)

Page	Description
Login Page	Role-based authentication
Student Dashboard	Upload + job summary
Upload Form	File + print settings
Job History	Status tracking with Job ID
Admin Panel	View all jobs, update status



Overall Experience

1. How It Started:

When I first joined SURE Trust, I only knew core Java—loops, arrays, OOP concepts. I had never built a web application. Spring Boot looked scary. I saw annotations like `@RestController`, `@Autowired`, `@Entity` and felt completely lost. I doubted if I could complete this project.

2. The Middle Phase:

The first two weeks were difficult. Setting up the project, connecting to MySQL, getting the first page to run—everything took longer than expected. Spring Security was the biggest challenge. I kept getting 403 errors. Login worked but pages wouldn't load. I spent 4 full days on it. I watched 3 YouTube playlists, read 10+ blog posts, and finally asked my mentor. When it finally worked, I felt like I had climbed a mountain.

File upload was another struggle. Files were saving with wrong paths on Windows. I learned about `File.separator` the hard way. But each error taught me something new.

3. Breakthrough Moments:

- First time login redirected to student dashboard → I took a screenshot and sent it to my family.
- First file uploaded successfully → I printed the file path on console and celebrated.
- First job status changed from Pending to Completed → I refreshed the page five times just to see it change.

These small victories kept me going.

4. How It Ended:

Today, when I run the application and see the login page, the dashboard, the job history—I feel proud. It is not perfect. The UI is basic. Some features are incomplete. But it works. I built this from nothing.

6. Gratitude:

I thank SURE Trust for giving me this opportunity. I thank my mentors T. Chandhana Ma'am and Pruthvi Raj Sir for their patience and guidance. They never made me feel stupid for asking basic questions. I also thank my fellow team members for their support.

7. Final Thought:

Before this project, I feared placement interviews because I had no real project to show. Now I have PAGE WAVE. I can confidently say: *"I built a Spring Boot application. I solved real problems. I can contribute to your team."*



Conclusion and Future Scope

Recap of Objectives & Achievements

The PAGE WAVE project was developed over 12 weeks to digitize manual printing systems in educational institutions. **All core objectives were successfully achieved:**

- ✓ Role-based authentication (Student/Admin)
- ✓ Document upload with validation (PDF/DOCX/JPG, 10MB limit)
- ✓ Print job configuration (color, copies, duplex, page range)
- ✓ Real-time job tracking with unique Job ID
- ✓ Student dashboard with personal print history
- ✓ Admin dashboard with job filtering and status updates
- ✓ Cost estimation (₹2 B&W, ₹5 Color per page)
- ✓ Secure session management

Pending Features: Email notifications, forgot password, real printer integration, payment gateway, pagination, mobile responsiveness.

Key Takeaways

- A complete web application was built using **zero-cost open-source technologies**
 - The system introduces **fairness and transparency** through a digital queue
 - Core functionality is **stable and demo-ready** for pilot testing
 - The project bridges the gap between **academic Java and industry-ready development**
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Future Scope

Short-Term (1-2 months):

- Pagination, forgot password, UI enhancement using Bootstrap
- Print preview, duplicate file prevention

Medium-Term (3-6 months):

- Email notifications, real printer integration (IPP/CUPS)
- UPI payment gateway, usage reports in PDF/Excel
- User profile management, analytics dashboard



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Long-Term (6-12 months):

- Cloud deployment (AWS/Azure/Railway)
- Mobile app (React Native/Flutter)
- QR code-based print release
- AI-based print scheduling, multilingual support

Final Word

PAGE WAVE started as a SURE Trust training project. It ended as a **portfolio-ready application** that solves a real problem. The system works. The learning is permanent. The scope for growth is immense.